Week 7

↳ NLP → Natural Language Processing

How are you → DL → I am fine

input — input — output

ANN's — Vectors (Num features) — Prob ($\hat{y}$) — "Text Data"

CNN's or GAN's — Images (Matrices) — Prob ($\hat{y}$)

NLP — Plain Text → Request — Plain text Response

H | W | A

Text data is Squence Data (Chain)

$x_1 \to x_2 \to x_3 \to x_4$

A | W | H

x_1 x_2 x_3 x_4 — I am fine

x_4 x_3 x_2 x_1 — fine am I

"Ethical"

NLP Objectives → Next Word Prediction / Completion
→ Text Sentiment Analysis
→ Q/A

I am → Indian
→ boy
→ girl
→ student

"Good food" → +ve
"Horrible food" → -ve

Terms

① Corpus :- All the text given to model for training
↓ Raw text
GPT-2 → Wikipedia

② Vocabulary :- List of unique words → tokens which the model learns

Sequence Modelling

Vocab = 4 '[I am a indian]'

I am → a / Indian → Indian

likelyhood

$P(a \mid I, am) > P(Indian \mid I, am)$

I am a

$P(I \mid I, am, a)$

Sq M → Objective

" $P(X_t \mid X_{t-1}, X_{t-2}, X_{t-3} \ldots)$ "

Model Training

- Preprocessing
- Model → RNN

Pre Processing

- Clean the corpus
  - IPL Page
  - "IPL is a t20 league"
- Learn Vocabulary = [IPL, is, a, t20, league]

↓

Vectorization

IPL = [1 0 0] = [0.3 0.5 0.1]

is = [0 1 0] = [0.98, 0.35, 0.67]

[0.5 0.6]

[0.71, 0.59]

Man Men

"Semantic Similarity"

Model (RNN) How are you doing

U = Input Matrix

U = input layer dim x h layer

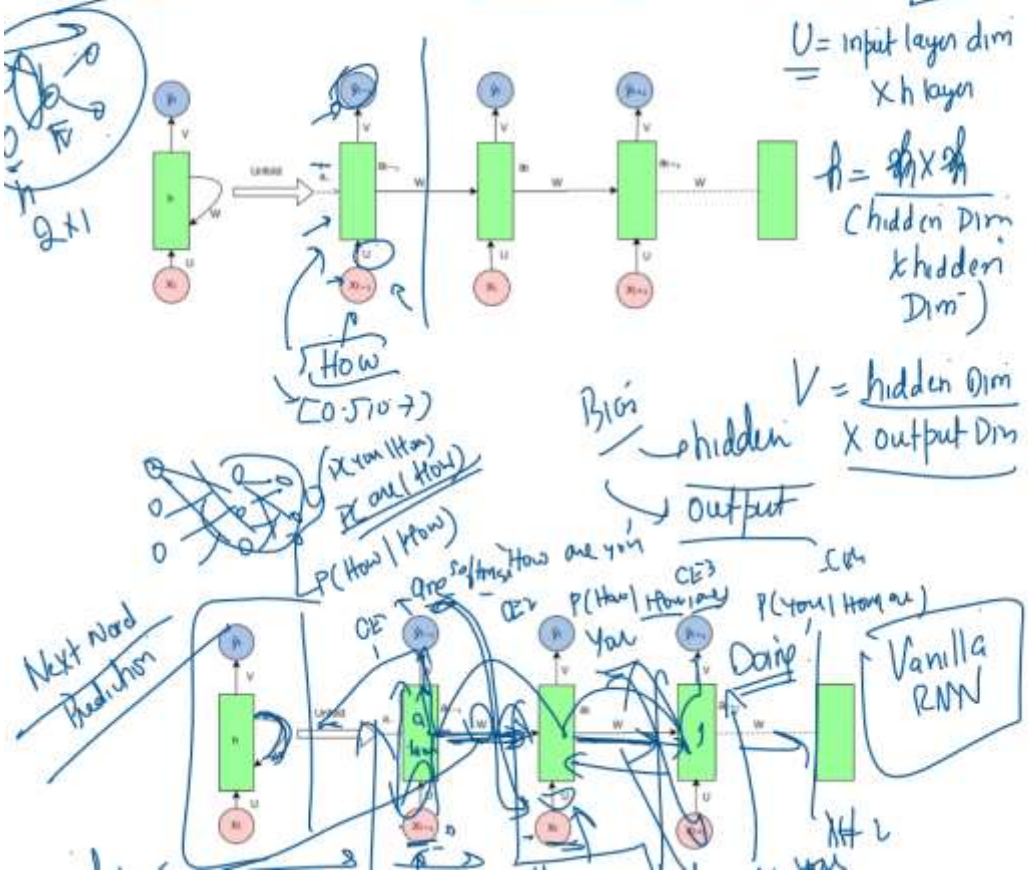
h = (hidden Dim x hidden Dim)

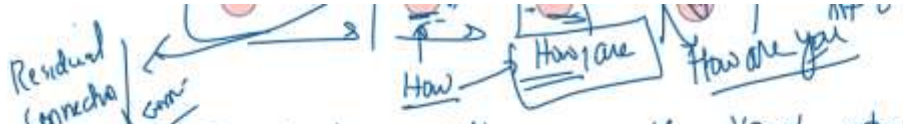
V = hidden Dim x output Dim

Bias

hidden

output





RNN'S do a softmax over the vocab after the output layer.

Params : U, V, H

Shared across timesteps

same for bias

weight

$U \rightarrow$ input matrix

$H \rightarrow$ Hidden WM (+ H_b bias)

$V \rightarrow$ Output WM (+ V_b bias)

$X_t =$ t^{th} word

$X_t =$ t^{th} word

$y_t =$ output at t^{th} timestep

loss

T_1	T_2	T_3	T_4
CE1	CE2	CE3	CE4

Model loss : Mean / Sum /

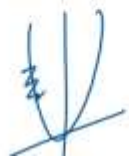
Issues in RNN

↳ Vanishing Gradients

$W_t = W_{t-1} - \eta \frac{\partial L}{\partial W_{t-1}}$

$W_t \approx W_{t-1} \rightarrow$ No update

LSTM / GRU



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gradient Clipping

Exploding Gradients

$$W_t = W_{t-1} - \eta \nabla_{W_{t-1}} L$$

$$W_t \approx W_{t-1}$$

Context Grabbing

New Delhi is the Capital of India

Bidirectional RNN